

Stability in Frequency domain

→ conditions for stability using Bode plot:

↳ $GM > 0\text{dB} \Rightarrow$ stable system

↳ $GM = 0\text{dB} \Rightarrow$ marginally stable system

↳ $GM < 0\text{dB} \Rightarrow$ unstable system

↳ $PM > 0^\circ \Rightarrow$ stable system

↳ $PM = 0^\circ \Rightarrow$ marginally stable system

↳ $PM < 0^\circ \Rightarrow$ unstable system

↳ $\omega_{pc} > \omega_{gc} \Rightarrow$ stable system

↳ $\omega_{pc} = \omega_{gc} \Rightarrow$ marginally stable system

↳ $\omega_{pc} < \omega_{gc} \Rightarrow$ unstable system

↳ $\omega_{gc} \Rightarrow$ gain crossover frequency $\Rightarrow$ frequency at which the open loop magnitude is 0dB (unity)

↳ $\omega_{pc} \Rightarrow$ phase crossover frequency $\Rightarrow$ frequency at which the open loop phase is 180°

↳ $GM \Rightarrow$ amount of gain reduction required to bring the system to verge of instability at the phase crossover frequency

↳ $PM \Rightarrow$ amount of gain reduction required to bring the system to verge of instability at the gain crossover frequency

→ conditions for stability using Nyquist plot:

↳ all closed loop poles must lie in the left-half s plane

$$N = P - Z$$

$$Z = 0 \Rightarrow N = P \Rightarrow \text{stable}$$

↳ $N \Rightarrow$ number of anticlockwise encirclement of critical point $(-1 + 0j)$ in complex plane

↳ $P \Rightarrow$ number of poles of open loop transfer function lying on right-half s-plane

↳ $Z \Rightarrow$ number of poles of closed loop transfer function lying on right-half s-plane

→ Encirclement of critical point $(-1 + 0j)$:

↳ no encirclement:

↳ no poles of $G(s)H(s)$ in right-half s plane $\Rightarrow$
 $P = 0 \Rightarrow \text{stable}$

↳ poles of $G(s)H(s)$ in right-half s plane $\Rightarrow P > 0$
 $\Rightarrow \text{unstable}$

↳ anticlockwise encirclement:

↳ number of anticlockwise encirclement =
number of poles of $G(s)H(s)$ in right-half s
plane $\Rightarrow N = P \Rightarrow \text{stable}$

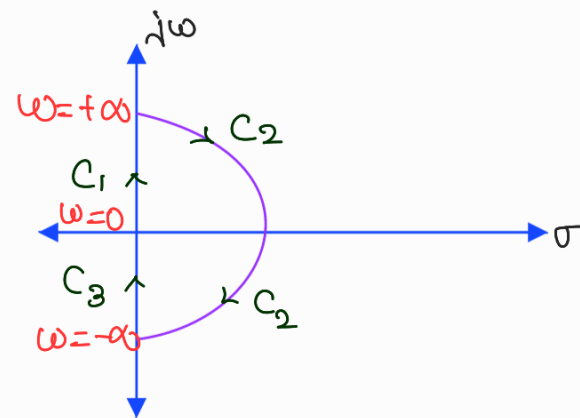
↳ number of anticlockwise encirclement $\neq$
number of poles of $G(s)H(s)$ in right-half s
plane $\Rightarrow N \neq P \Rightarrow \text{unstable}$

↳ clockwise encirclement:

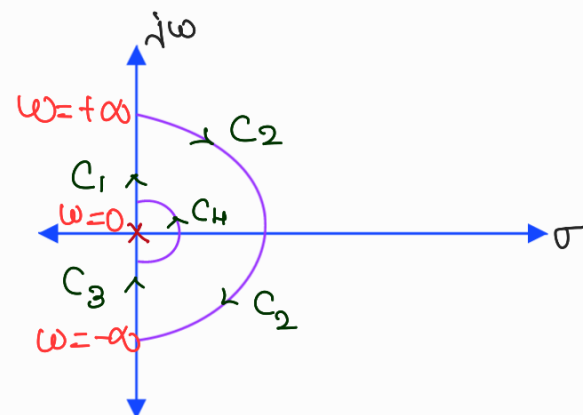
↳ number of clockwise encirclement =
number of poles of $G(s)H(s)$ in right-half s
plane $\Rightarrow$ always unstable

→ Nyquist contour:

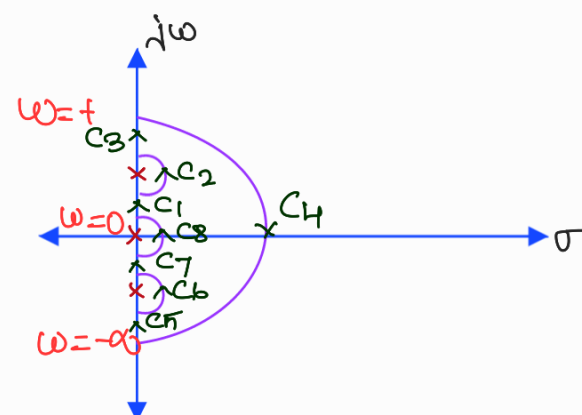
↳ when there is no pole on
imaginary axis:



↳ when there are poles at
origin:



↳ when there are poles on
imaginary axis and at
origin:



Polar plot

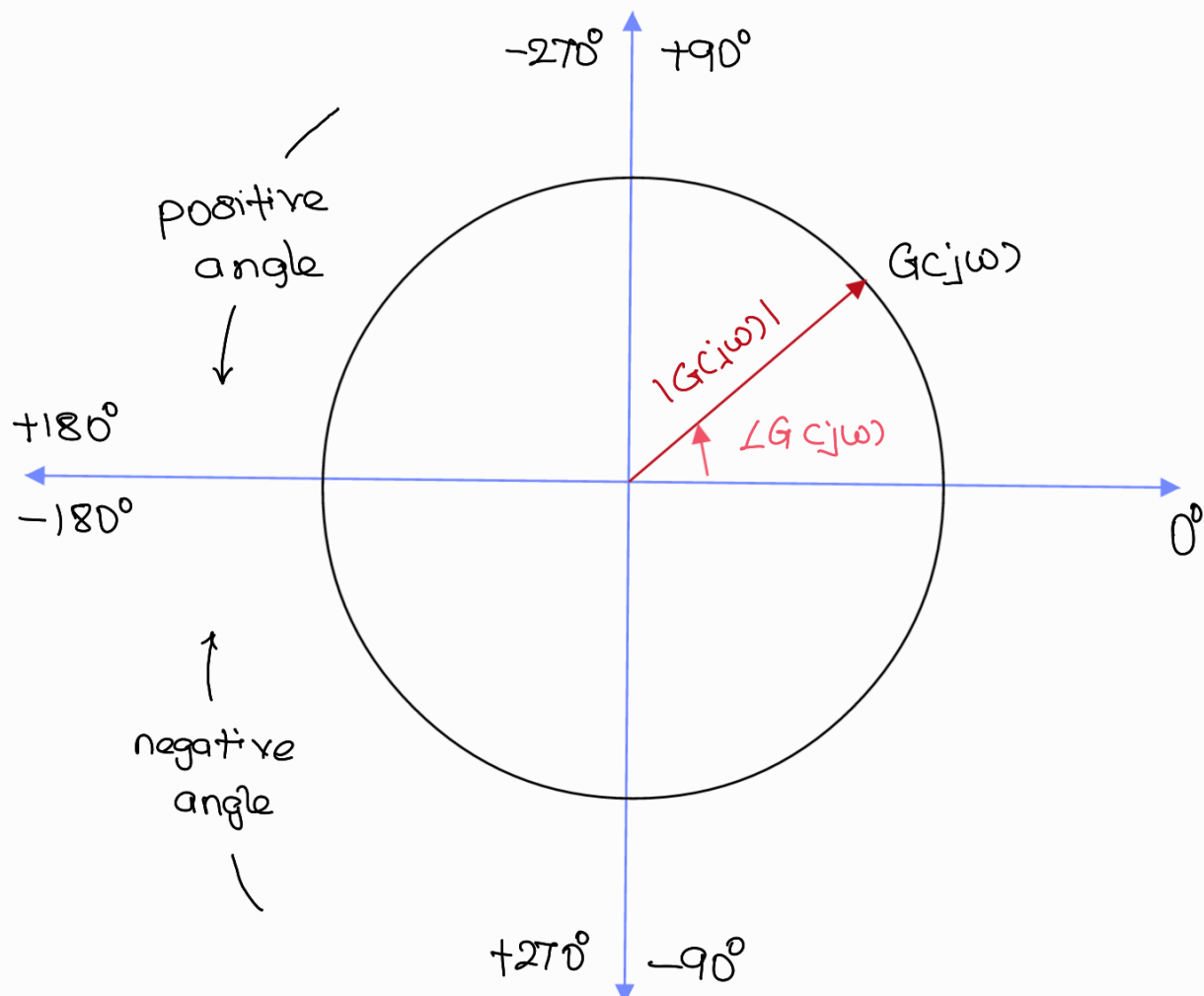
→ plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity.

→ locus of vectors of $|G(j\omega)| \angle G(j\omega)$ as ω is varied from zero to infinity.

→ polar plot:

↳ concentric circle → magnitude

↳ radial lines → phase angles



1. The open loop transfer function of a unity feedback system is given by $G(s)$. sketch the polar plot and determine the gain margin and phase margin.

$$G(s) = \frac{1}{s(1+s)(1+2s)}$$

$$s \rightarrow j\omega$$

$$G(j\omega) = \frac{1}{j\omega(1+j\omega)(1+2j\omega)}$$

corner frequency:

$$\omega_{c1} = \frac{1}{2} = 0.5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{1} = 1 \text{ rad/sec}$$

magnitude:

$$|G(j\omega)| = \frac{1}{\omega \sqrt{1+\omega^2} \sqrt{1+4\omega^2}}$$

$$= \frac{1}{\omega \sqrt{(1+\omega^2)} \sqrt{1+4\omega^2}}$$

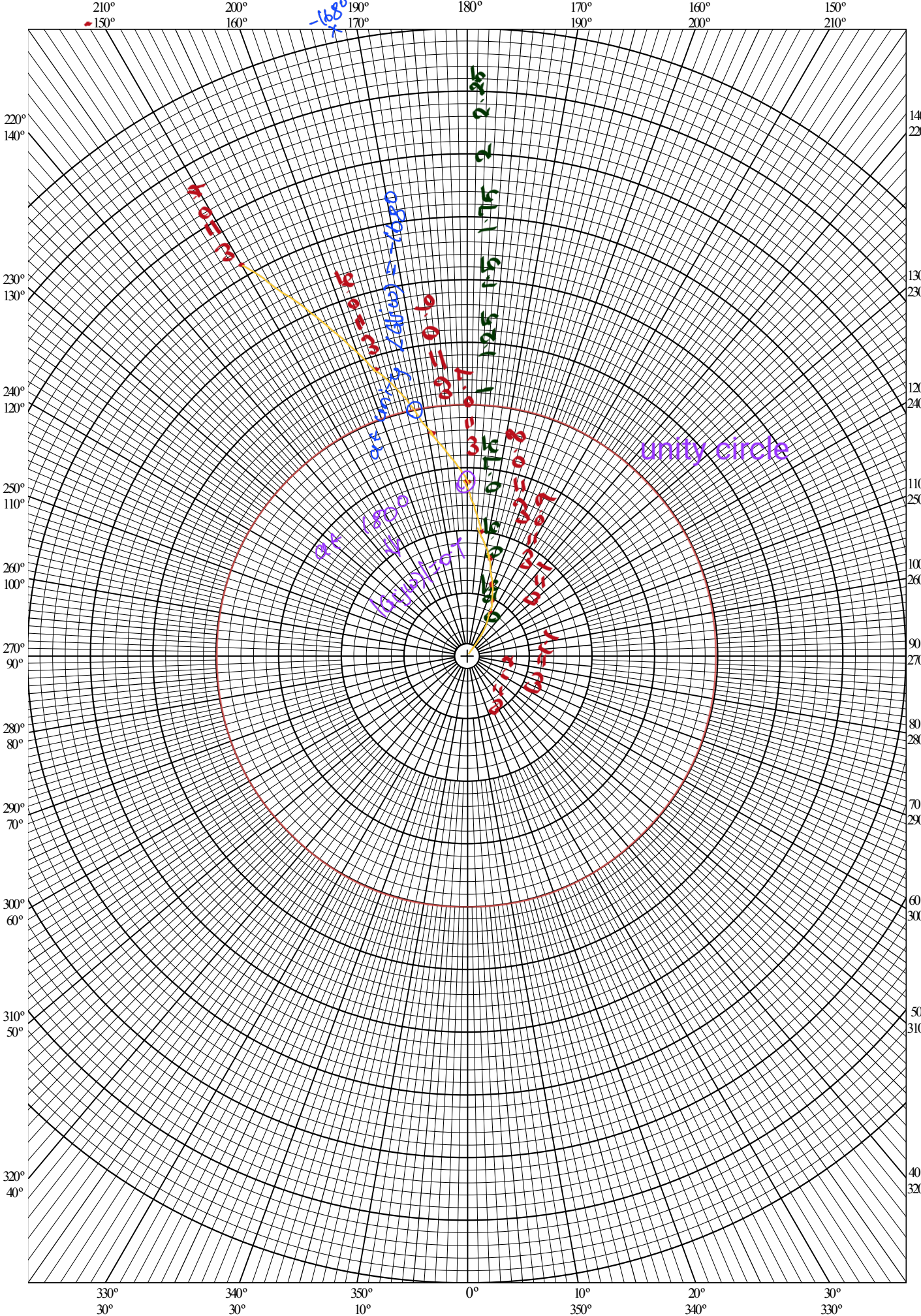
$$= \frac{1}{\omega \sqrt{(1+\omega^2)(1+4\omega^2)}}$$

$$|G(j\omega)| = \frac{1}{\omega \sqrt{1+5\omega^2+4\omega^4}}$$

phase:

$$\angle G(j\omega) = -90^\circ - \tan^{-1} \omega - \tan^{-1} 2\omega$$

ω	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2
$ G(j\omega) $	1.8	1.2	0.9	0.7	0.5	0.4	0.3	0.25	0.2
$\angle G(j\omega)$	-150°	-162°	-171°	-180°	-187°	-193°	-198°	-203°	-207°



GM:

$$k_g = \frac{1}{|G_C(j\omega)|}$$

$$= \frac{1}{0.7}$$

PM:

$$k_g = 1.43$$

$$\gamma = 180^\circ + \phi_{gc}$$

$$= 180^\circ - 168^\circ$$

$$\gamma = 12^\circ$$

→ rectangular coordinates: $G(j\omega) = G_R(j\omega) + jG_I(j\omega)$

ω	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2
$G_R(j\omega)$	-1.56	-1.14	-0.89	-0.7	-0.5	-0.39	-0.29	-0.23	-0.18
$G_I(j\omega)$	-0.9	-0.37	-0.14	0	0.06	0.08	0.09	0.1	0.09

GM:

$$k_g = \frac{1}{|G_C(j\omega)|}$$

$$= \frac{1}{0.7}$$

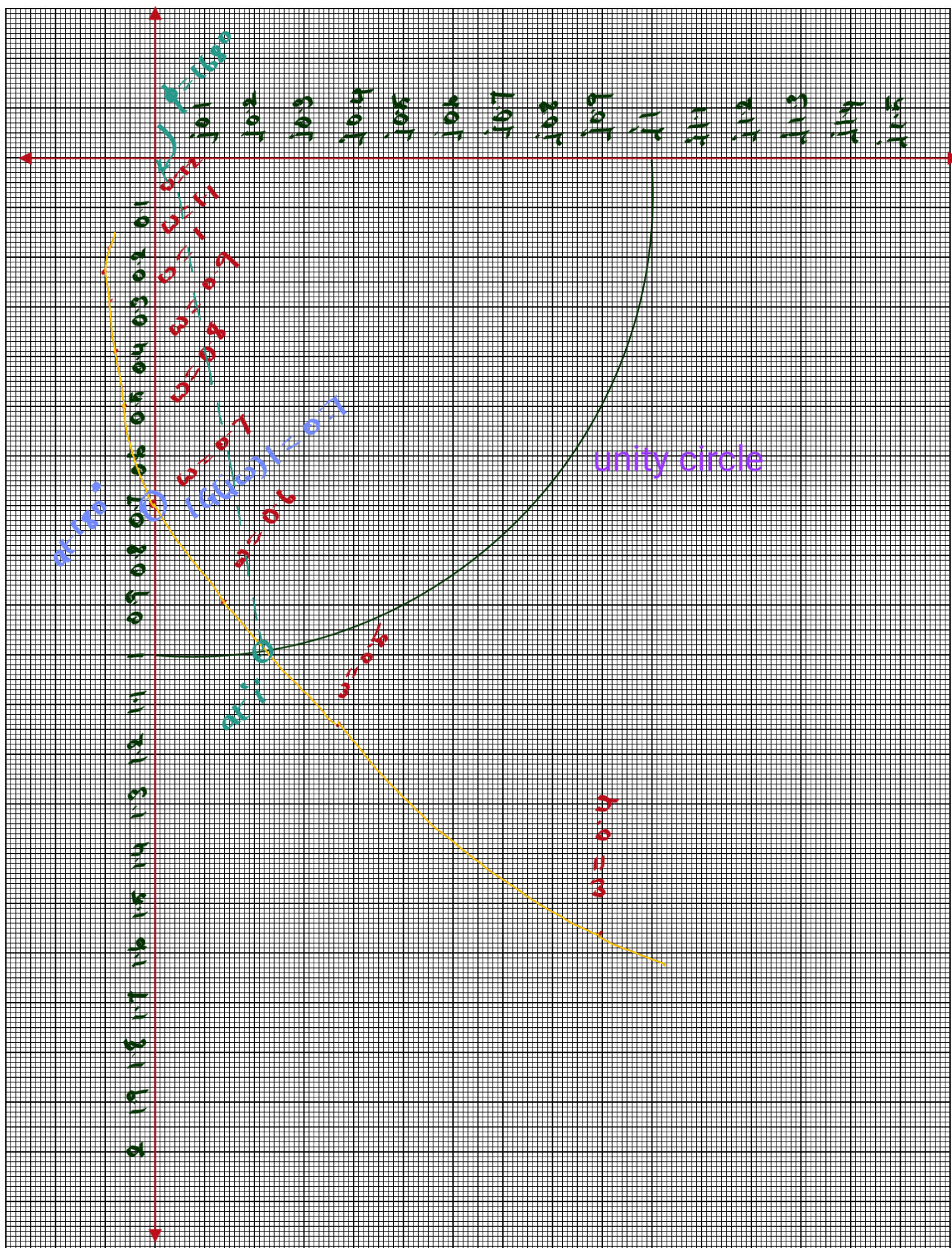
PM:

$$k_g = 1.43$$

$$\gamma = 180^\circ + \phi_{gc}$$

$$= 180^\circ - 168^\circ$$

$$\gamma = 12^\circ$$



2. The open loop transfer function of a unity feedback system is given by $G(s)$. sketch the polar plot and determine the gain margin and phase margin.

$$G(s) = \frac{1}{8C(1+s)^2}$$

$$s \rightarrow j\omega$$

$$G(j\omega) = \frac{1}{j\omega C(1+j\omega)^2}$$

corner frequency:

$$\omega_{c1} = \frac{1}{1} = 1 \text{ rad/sec}$$

magnitude:

$$|G(j\omega)| = \frac{1}{\omega \sqrt{1^2 + \omega^2} \sqrt{1^2 + \omega^2}}$$

$$= \frac{1}{\omega \sqrt{1+\omega^2} \sqrt{1+\omega^2}}$$

$$= \frac{1}{\omega \sqrt{(1+\omega^2)^2}} = \frac{1}{\omega (1+\omega^2)}$$

$$|G(j\omega)| = \frac{1}{\omega + \omega^3}$$

phase:

$$\angle G(j\omega) = -90^\circ - 2 \tan^{-1} \omega$$

ω	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2
$ G(j\omega) $	2.2	1.6	1.2	1	0.8	0.6	0.5	0.4	0.3
$\angle G(j\omega)$	-134°	-148°	-161°	-169°	-167°	-174°	-180°	-185°	-190°

GM:

$$K_g = \frac{1}{|G(j\omega)|}$$

$$= \frac{1}{0.5}$$

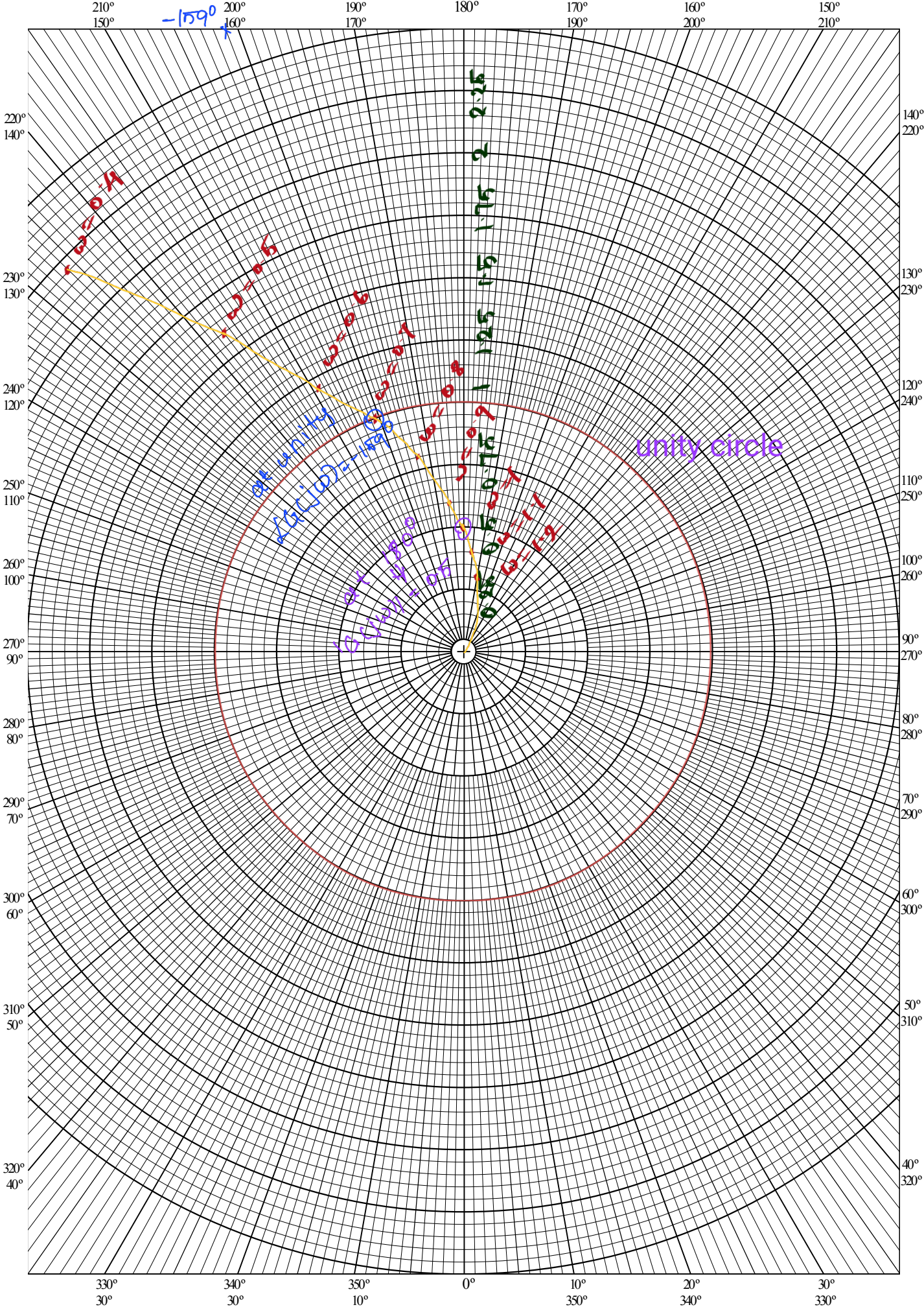
$$K_g = 2$$

PM:

$$P = 180^\circ + \phi_{gc}$$

$$= 180^\circ - 159^\circ$$

$$P = 21^\circ$$



→ rectangular coordinates: $G(j\omega) = G_R(j\omega) + jG_I(j\omega)$

ω	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2
$G_R(j\omega)$	-1.53	-1.28	-1.05	-0.93	-0.78	-0.6	-0.5	-0.4	-0.3
$G_I(j\omega)$	-1.58	-0.96	-0.58	-0.36	-0.18	0.06	0	0.03	0.05

GM:

$$k_g = \frac{1}{|G(j\omega)|}$$

$$= \frac{1}{0.5}$$

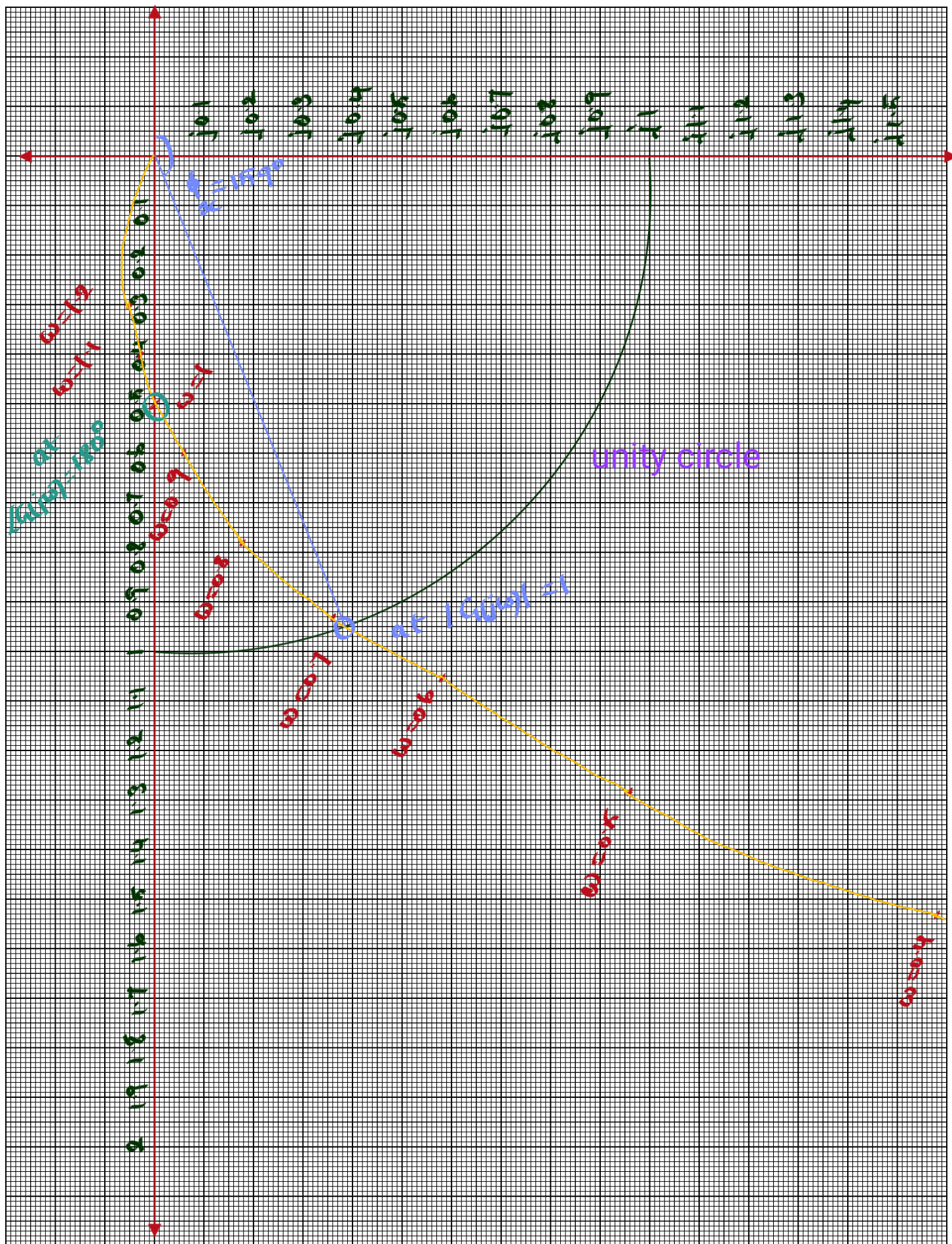
$$k_g = 2$$

PM:

$$\gamma = 180^\circ + \phi_{gc}$$

$$= 180^\circ - 159^\circ$$

$$\gamma = 21^\circ$$



3. The open loop transfer function of a unity feedback system is given by $G(s)$. sketch the polar plot and determine the gain margin and phase margin.

$$G(s) = \frac{1}{s^2 (1+s) (1+2s)}$$

$$s \rightarrow j\omega$$

$$G(j\omega) = \frac{1}{(j\omega)^2 (1+j\omega) (1+2j\omega)}$$

corner frequency:

$$\omega_{c1} = \frac{1}{2} = 0.5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{1} = 1 \text{ rad/sec}$$

magnitude:

$$|G(j\omega)| = \frac{1}{\omega^2 \sqrt{1+\omega^2} \sqrt{1+4\omega^2}}$$

$$= \frac{1}{\omega^2 \sqrt{(1+\omega^2)} \sqrt{1+4\omega^2}}$$

$$= \frac{1}{\omega^2 \sqrt{(1+\omega^2) (1+4\omega^2)}}$$

$$|G(j\omega)| = \frac{1}{\omega^2 \sqrt{1+5\omega^2+4\omega^4}}$$

phase:

$$\angle G(j\omega) = -180^\circ - \tan^{-1} \omega - \tan^{-1} 2\omega$$

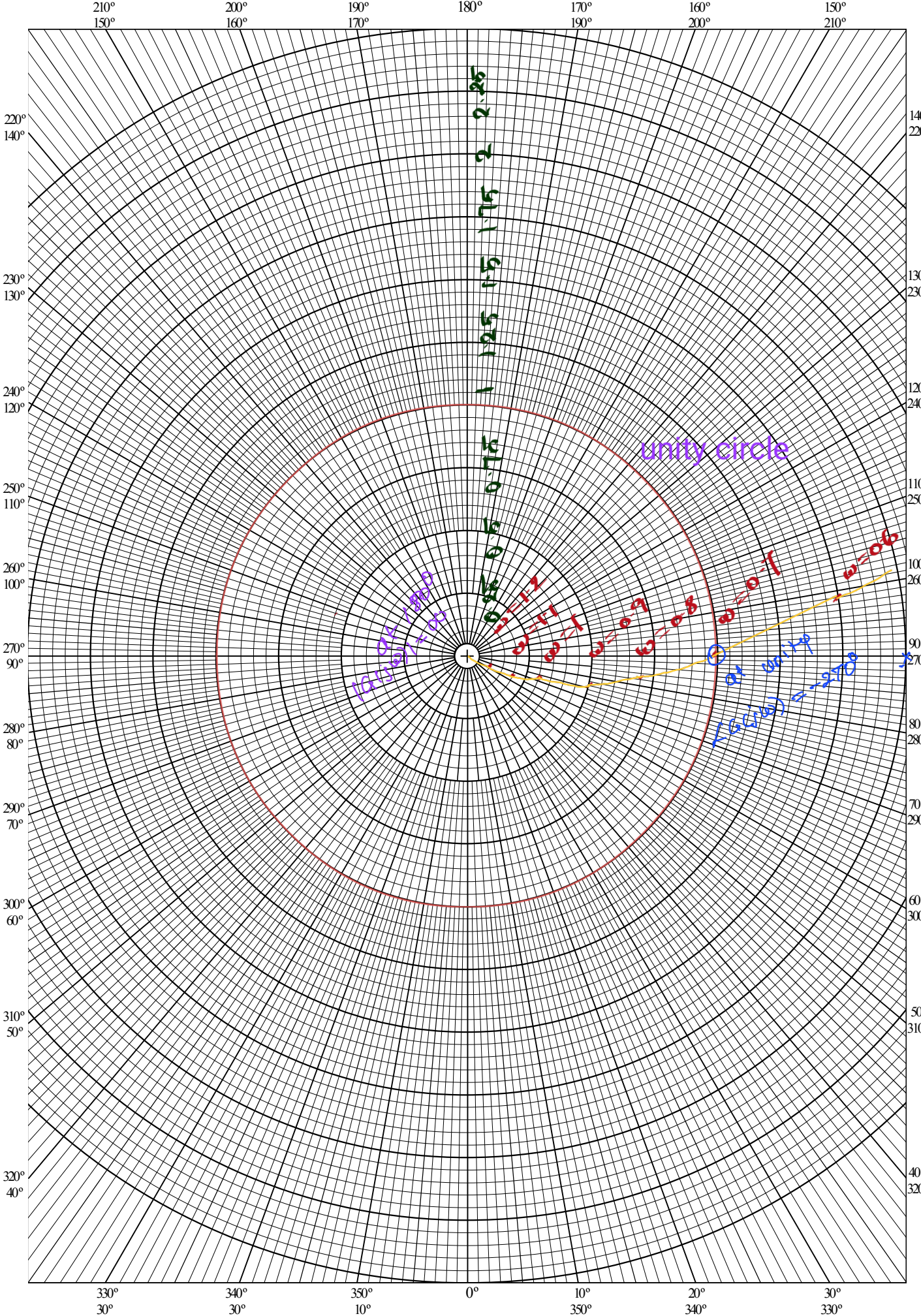
ω	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2
$ G(j\omega) $	4.5	2.5	1.5	0.97	0.7	0.5	0.3	0.2	0.1
$\angle G(j\omega)$	-240°	-251°	-261°	-269°	-277°	-283°	-288°	-293°	-298°

GM:

$$K_g = \frac{1}{|G(j\omega)|} = \frac{1}{\infty} = 0$$

PM:

$$P = 180^\circ + \phi_{gc} = 180^\circ - 270^\circ = -90^\circ$$



4. The open loop transfer function of a unity feedback system is given by $G(s)$. sketch the polar plot and determine the gain margin and phase margin.

$$G(s) = \frac{(1+0.2s)(1+0.02\pi s)}{s^3(1+0.005s)(1+0.001s)}$$

BODE PLOT

→ frequency response plot of the sinusoidal transfer function of the system.

→ parts:

↳ magnitude plot $\Rightarrow$ magnitude of a sinusoidal transfer function vs frequency.

$$20 \log_{10} |G(j\omega)| \text{ vs } \log \omega$$

↳ phase plot $\Rightarrow$ phase angle of a sinusoidal transfer function vs frequency.

$$\angle G(j\omega) \text{ vs } \log \omega$$

→ usually drawn for open loop system.

→ merits:

↳ simplest method

↳ multiplication of magnitude can be converted into addition.

↳ division of magnitude can be converted into subtraction.

↳ low and high frequency characteristics of transfer function can be determined.

basic factors of open loop transfer function $G(j\omega)$:

→ constant gain:

$$G(s) = k$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = k = k \angle 0^\circ$$

$$A = |G(j\omega)| = 20 \log k$$

$$\phi = \angle G(j\omega) = 0^\circ$$

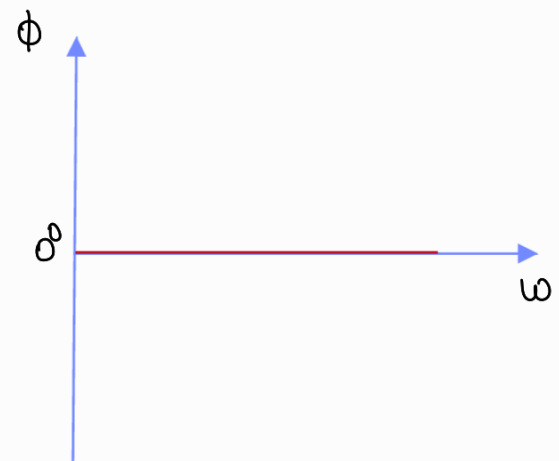
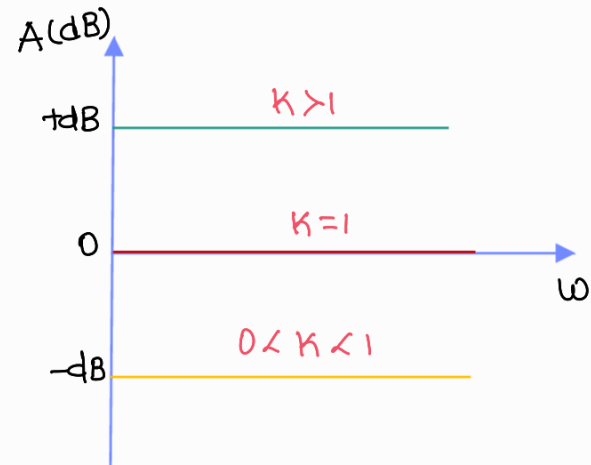
magnitude plot:

$$k=1 \Rightarrow A = 20 \log 1 = 0 \text{ dB}$$

$$k > 1 \Rightarrow A = + \text{dB}$$

$$0 < k < 1 \Rightarrow A = - \text{dB}$$

phase plot: $\phi = 0^\circ$



→ integral factor:

$$G(s) = \frac{k}{s}$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = \frac{k}{j\omega} = \frac{k}{\omega} \angle -90^\circ$$

$$A = |G(j\omega)| = 20 \log \frac{k}{\omega}$$

$$\phi = \angle G(j\omega) = -90^\circ$$

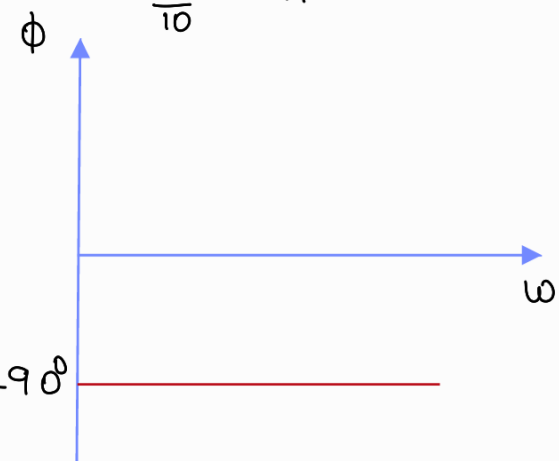
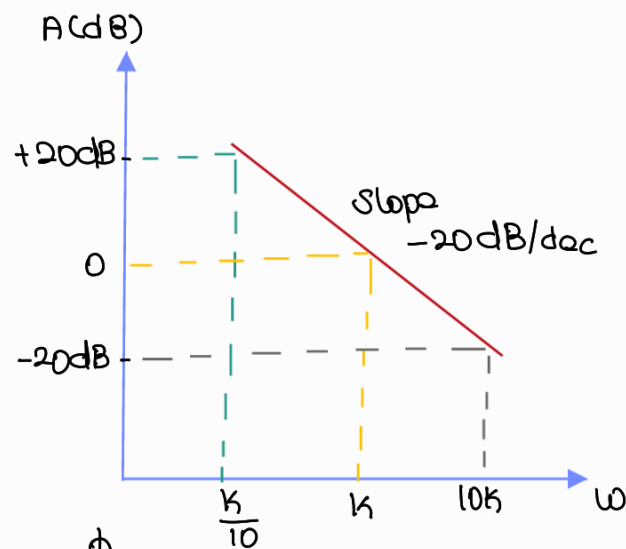
magnitude plot:

$$\omega = \frac{k}{10} \Rightarrow A = 20 \log 10 = 20 \text{ dB}$$

$$\omega = k \Rightarrow A = 20 \log 1 = 0 \text{ dB}$$

$$\omega = 10k \Rightarrow A = 20 \log \left(\frac{1}{10}\right) = -20 \text{ dB}$$

phase plot: $\phi = -90^\circ$



→ derivative factor:

$$G(s) = k s$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = k(j\omega) = k\omega \angle +90^\circ$$

$$A = |G(j\omega)| = 20 \log(k\omega)$$

$$\phi = \angle G(j\omega) = +90^\circ$$

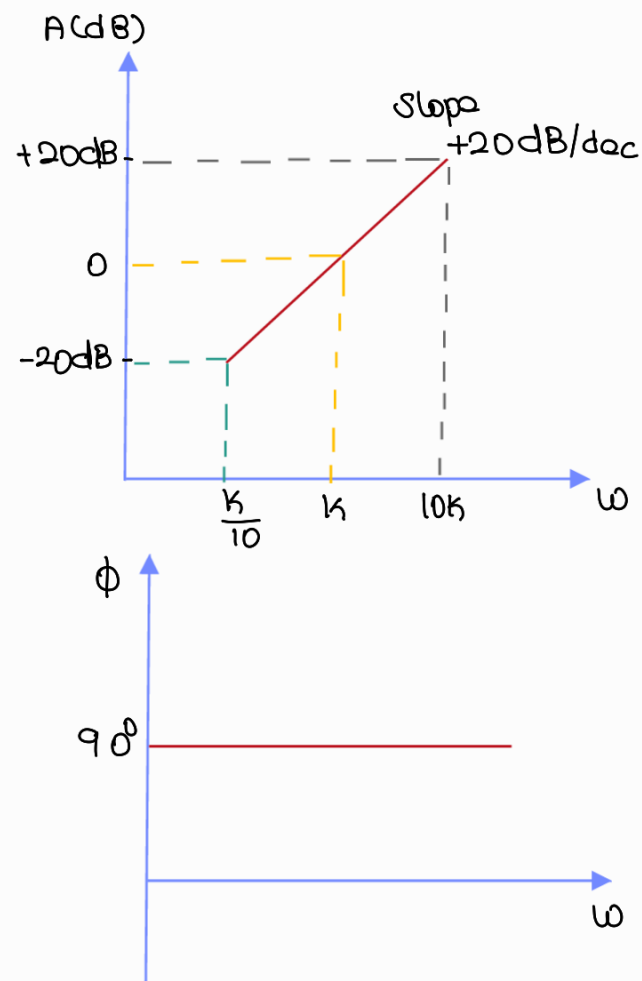
magnitude plot:

$$\omega = \frac{k}{10} \Rightarrow A = 20 \log(10) = -20 \text{ dB}$$

$$\omega = k \Rightarrow A = 20 \log 1 = 0 \text{ dB}$$

$$\omega = 10k \Rightarrow A = 20 \log 10 = +20 \text{ dB}$$

phase plot: $\phi = +90^\circ$



→ first order factor in denominator:

$$G(s) = \frac{1}{1+sT}$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = \frac{1}{1+j\omega T} = \frac{1}{\sqrt{1+\omega^2 T^2}} \angle -\tan^{-1}\omega T$$

$$A = 20 \log \left(\frac{1}{\sqrt{1+\omega^2 T^2}} \right)$$

$$A = -20 \log \sqrt{1+\omega^2 T^2}$$

magnitude plot:

low frequency $\Rightarrow \omega T \ll 1$

$$A = -20 \log 1 = 0 \text{ dB}$$

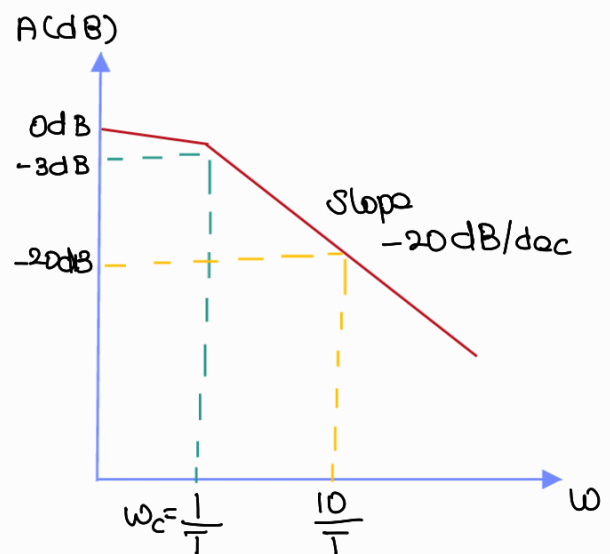
$$\omega = \omega_c = \frac{1}{T} \Rightarrow A = -20 \log \sqrt{1+1} = -3 \text{ dB}$$

↳ corner frequency/ break frequency

high frequency $\Rightarrow \omega T \gg 1$

$$A = -20 \log \omega T$$

$$\omega = \frac{10}{T} \Rightarrow A = -20 \log 10 = -20 \text{ dB}$$



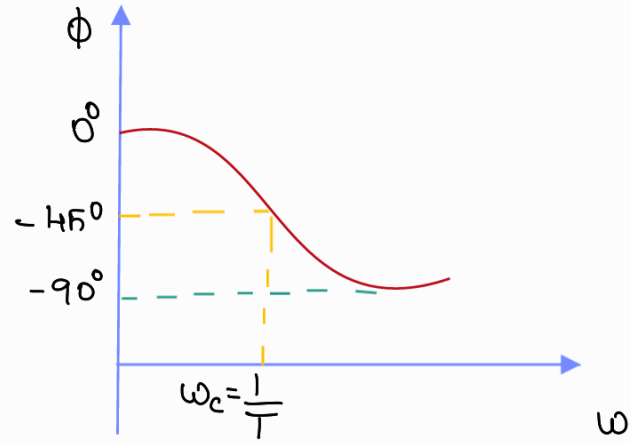
$$\phi = -\tan^{-1} \omega T$$

phase plot:

$$\omega = 0 \Rightarrow \phi = -\tan^{-1} 0 = 0^\circ$$

$$\omega = \omega_c = \frac{1}{T} \Rightarrow \phi = -\tan^{-1} 1 = -45^\circ$$

$$\omega = \infty \Rightarrow \phi = -\tan^{-1} \infty = -90^\circ$$



→ first order factor in numerator:

$$G(s) = 1 + sT$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = 1 + j\omega T = \sqrt{1 + \omega^2 T^2} \angle \tan^{-1} \omega T$$

$$A = 20 \log \sqrt{1 + \omega^2 T^2}$$

$$\phi = \tan^{-1} \omega T$$

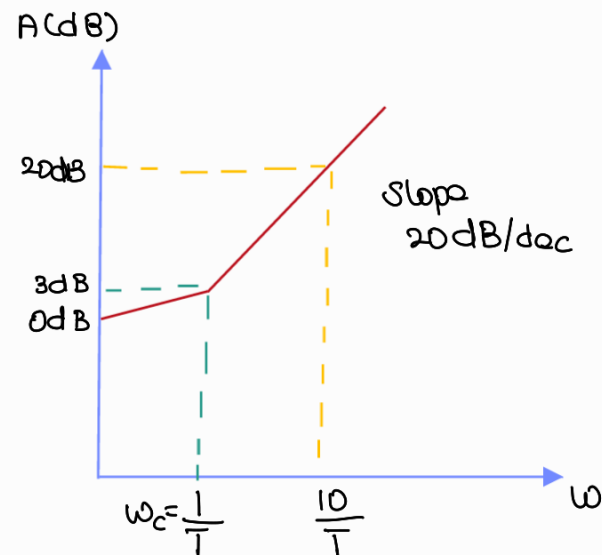
magnitude plot:

low frequency $\Rightarrow \omega T \ll 1$

$$A = -20 \log 1 = 0 \text{ dB}$$

$$\omega = \omega_c = \frac{1}{T} \Rightarrow A = 20 \log \sqrt{1+1} = 3 \text{ dB}$$

↳ corner frequency / break frequency



high frequency $\Rightarrow \omega T \gg 1$

$$A = 20 \log \omega T$$

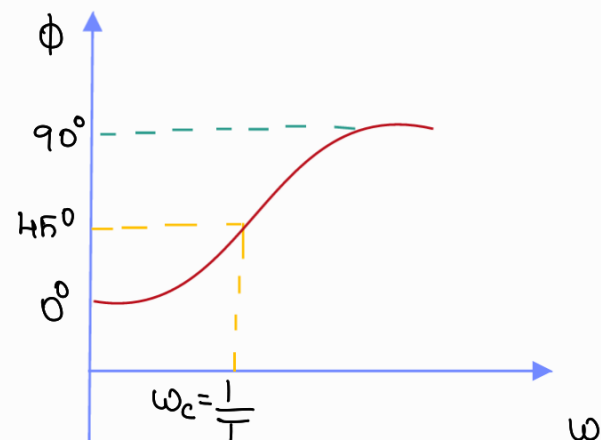
$$\omega = \frac{10}{T} \Rightarrow A = 20 \log 10 = 20 \text{ dB}$$

phase plot:

$$\omega = 0 \Rightarrow \phi = \tan^{-1} 0 = 0^\circ$$

$$\omega = \omega_c = \frac{1}{T} \Rightarrow \phi = \tan^{-1} 1 = 45^\circ$$

$$\omega = \infty \Rightarrow \phi = \tan^{-1} \infty = 90^\circ$$



→ quadratic factor in denominator:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(s) = \frac{1}{1 + 2\zeta \frac{s}{\omega_n} + \left(\frac{s}{\omega_n}\right)^2}$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = \frac{1}{1 + 2\zeta \frac{j\omega}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2}$$

$$G(j\omega) = \frac{1}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + j2\zeta \frac{\omega}{\omega_n}}$$

$$G(j\omega) = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}} \angle -\tan^{-1} \frac{2\zeta \frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}$$

$$A = 20 \log \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}}$$

$$= -20 \log \sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}$$

$$= -20 \log \sqrt{1 + \frac{\omega^4}{\omega_n^4} - 2\frac{\omega^2}{\omega_n^2} + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}$$

$$A = -20 \log \sqrt{1 - \frac{\omega^2}{\omega_n^2} (2 - 4\zeta^2) + \frac{\omega^4}{\omega_n^4}}$$

low frequency $\Rightarrow \omega \ll \omega_n$

$$A = -20 \log 1 = 0 \text{ dB}$$

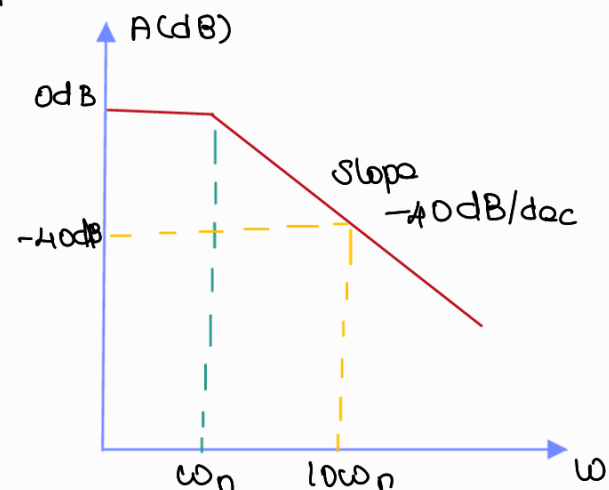
high frequency $\Rightarrow \omega \gg \omega_n$

$$A = -20 \log \sqrt{\frac{\omega^4}{\omega_n^4}} = -20 \log \left(\frac{\omega}{\omega_n}\right)^2$$

$$A = -40 \log \frac{\omega}{\omega_n}$$

$$\omega = \omega_n \Rightarrow A = -40 \log 1 = 0 \text{ dB}$$

$$\omega = 10\omega_n \Rightarrow A = -40 \log 10 = -40 \text{ dB}$$

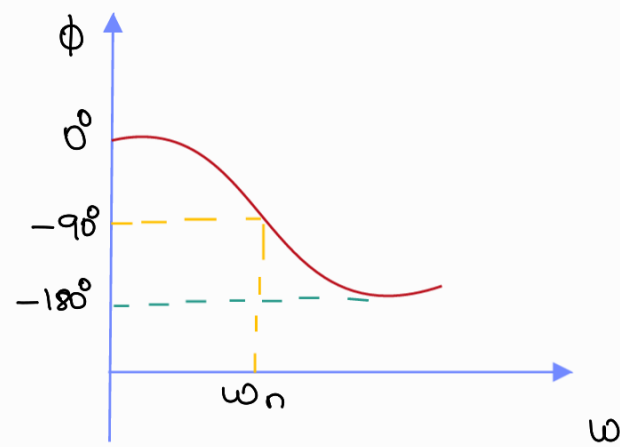


phase plot:

$$\omega = 0 \Rightarrow \phi = -\tan^{-1} 0 = 0^\circ$$

$$\omega = \omega_n \Rightarrow \phi = -\tan^{-1} \infty = -90^\circ$$

$$\omega = \infty \Rightarrow \phi = -\tan^{-1} 0 = -180^\circ$$



→ quadratic factor in numerator:

$$G(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

$$G(s) = 1 + 2\zeta \frac{s}{\omega_n} + \left(\frac{s}{\omega_n}\right)^2$$

$$s \rightarrow j\omega \Rightarrow G(j\omega) = 1 + \frac{2\zeta j\omega}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2$$

$$G(j\omega) = 1 + \left(\frac{j\omega}{\omega_n}\right)^2 + j2\zeta \frac{\omega}{\omega_n}$$

$$G(j\omega) = \sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}} \angle \tan^{-1} \frac{2\zeta \frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}$$

$$A = 20 \log \sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}$$

$$= 20 \log \sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}$$

$$= 20 \log \sqrt{1 + \frac{\omega^4}{\omega_n^4} - 2\frac{\omega^2}{\omega_n^2} + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}$$

$$A = 20 \log \sqrt{1 - \frac{\omega^2}{\omega_n^2}(2 - 4\zeta^2) + \frac{\omega^4}{\omega_n^4}}$$

low frequency $\Rightarrow \omega \ll \omega_n$

$$A = 20 \log 1 = 0 \text{ dB}$$

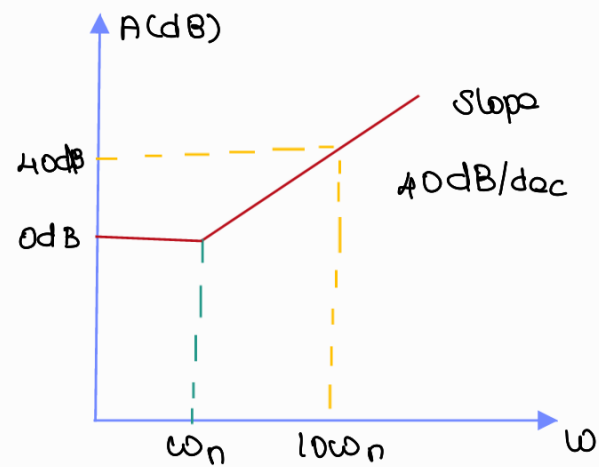
high frequency $\Rightarrow \omega \gg \omega_n$

$$A = 20 \log \sqrt{\frac{\omega^4}{\omega_n^4}} = -20 \log \left(\frac{\omega}{\omega_n} \right)^2$$

$$A = 40 \log \frac{\omega}{\omega_n}$$

$$\omega = \omega_n \Rightarrow A = 40 \log 1 = 0 \text{ dB}$$

$$\omega = 10\omega_n \Rightarrow A = 40 \log 10 = 40 \text{ dB}$$

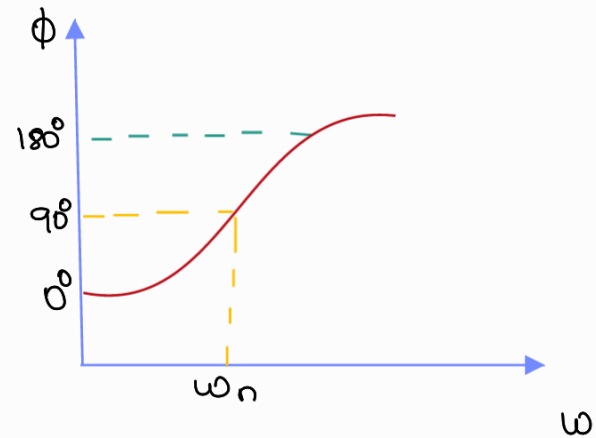


phase plot:

$$\omega = 0 \Rightarrow \phi = \tan^{-1} 0 = 0^\circ$$

$$\omega = \omega_n \Rightarrow \phi = \tan^{-1} 1 = 90^\circ$$

$$\omega = \infty \Rightarrow \phi = \tan^{-1} \infty = 180^\circ$$



→ magnitude plot:

↳ convert the transfer function into Bode form.

$$G(s) = \frac{K(1+sT_1)}{s(1+sT_2)\left[1+\frac{s^2}{\omega_n^2}+2\zeta\frac{s}{\omega_n}\right]} \xrightarrow{s \rightarrow j\omega} G(j\omega) = \frac{K(1+j\omega T_1)}{j\omega(1+j\omega T_2)\left[1-\frac{\omega^2}{\omega_n^2}+2j\zeta\frac{\omega}{\omega_n}\right]}$$

↳ list the corner frequencies in the increasing order.

$$G(j\omega) = \frac{K(1+j\omega T_1)}{j\omega(1+j\omega T_2)(1+j\omega T_3)} \quad ; \quad \text{let } T_3 < T_2 < T_1$$

corner frequencies:

$$\omega_{c1} = \frac{1}{T_1}, \omega_{c2} = \frac{1}{T_2}, \omega_{c3} = \frac{1}{T_3} \quad ; \quad \omega_{c1} < \omega_{c2} < \omega_{c3}$$

Term	corner frequency rad/sec	slope dB/sec	change in slope dB/sec
$j\omega$	—		
$(1 + j\omega T_1)$	ω_{c1}		
$1/(1 + j\omega T_2)$	ω_{c2}		
$1/(1 + j\omega T_3)$	ω_{c3}		

↳ choose an arbitrary frequency ω_l which is lesser than lowest corner frequency.

$$\omega_l < \omega_{c1}$$

↳ calculate the magnitude in dB of first term at arbitrary frequency and lowest corner frequency.

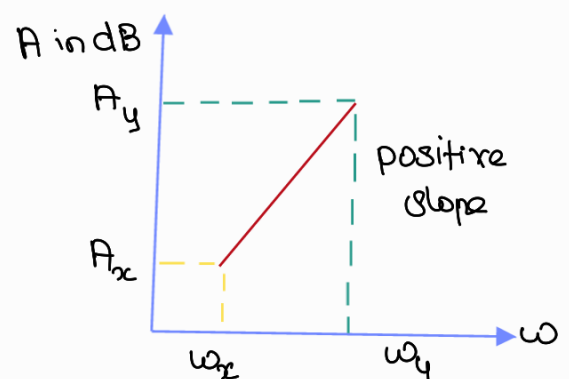
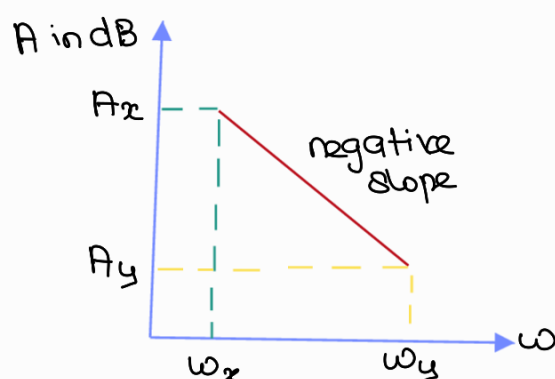
$$\omega = \omega_l \Rightarrow A = 20 \log |j\omega|$$

$$\omega = \omega_{c1} \Rightarrow A = 20 \log |j\omega|$$

↳ calculate the magnitude in dB at every corner frequency.

$$A_{\text{at } \omega_y} = \text{change in gain from } \omega_x \text{ to } \omega_y + \text{Gain at } \omega_x$$

$$A_{\text{at } \omega_y} = \left[\text{slope from } \omega_x \text{ to } \omega_y \times \log \frac{\omega_y}{\omega_x} \right] + \text{Gain at } \omega_x$$



↳ choose an arbitrary frequency ω_h which is higher than highest corner frequency.

$$\omega_{c3} < \omega_h$$

↳ calculate the magnitude in dB at arbitrary frequency.

$$A_{\text{at } \omega_h} = \text{change in gain from } \omega_{c3} \text{ to } \omega_h + \text{Gain at } \omega_{c3}$$

$$A_{\text{at } \omega_h} = \left[\text{slope from } \omega_{c3} \text{ to } \omega_h * \log \frac{\omega_h}{\omega_{c3}} \right] + \text{Gain at } \omega_{c3}$$

↳ mark all points in a semilog graph sheet and join the points by straight line.

→ phase plot :

↳ compute the phase angles of $G(j\omega)$ for various values of corner frequencies.

ω rad/sec	$\phi = \angle G(j\omega)$ deg
ω_L ω_{c1} ω_{c2} ω_{c3} ω_h	

→ gain margin and phase margin :

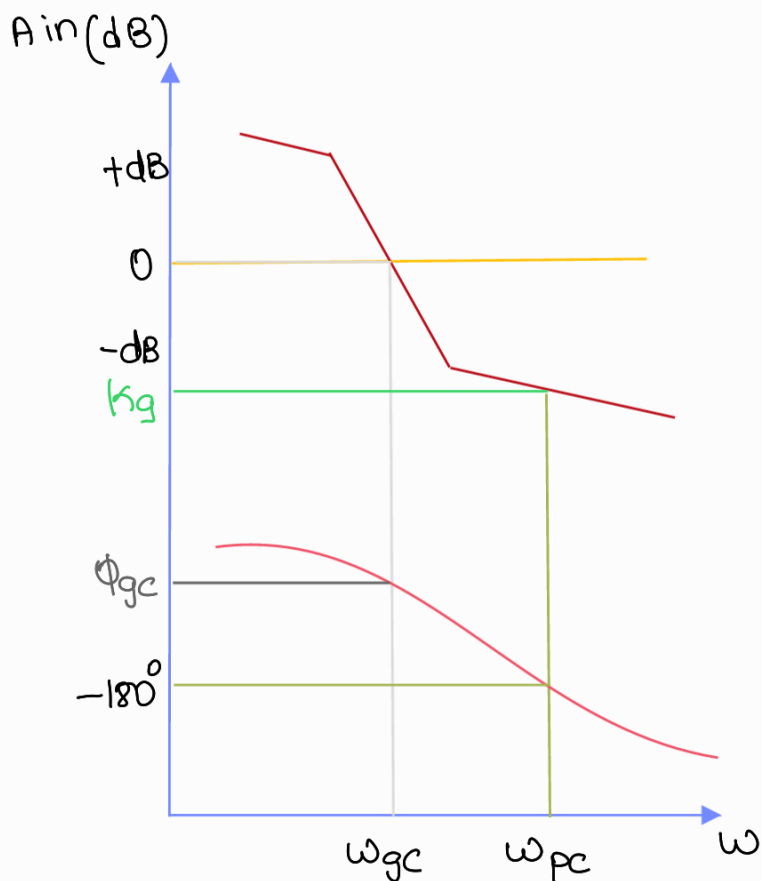
$$k_g = -20 \log |G(j\omega_{pc})|$$

$$\omega_{pc} = \omega \text{ at } \angle G(j\omega) = 180^\circ$$

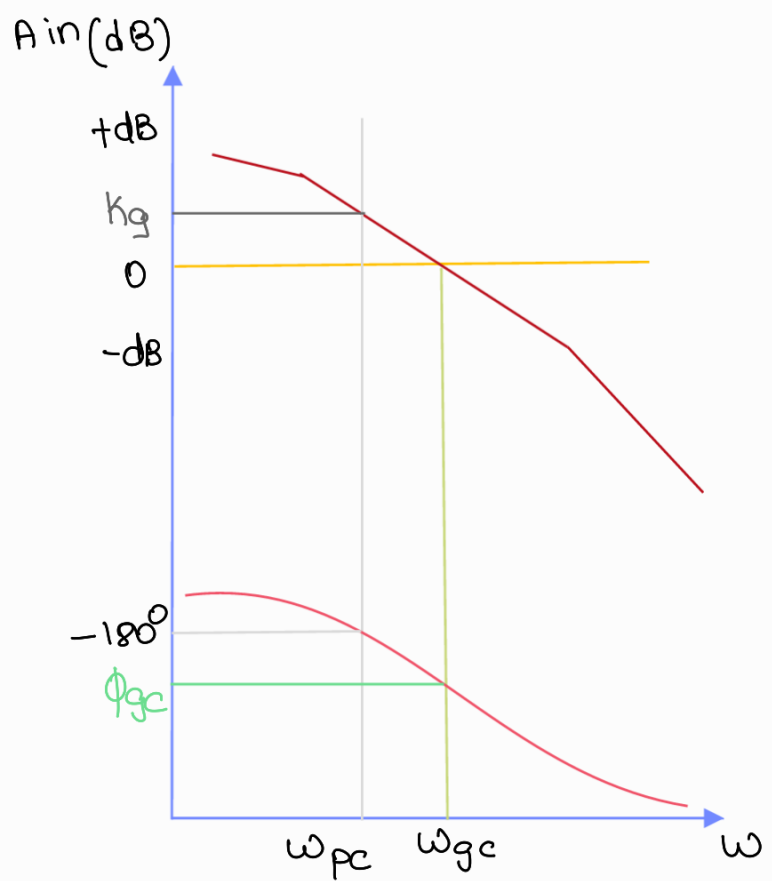
$$\gamma = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = \angle G(j\omega_{gc})$$

$$\omega_{gc} = \omega \text{ at } |G(j\omega)| = 1 \text{ in dB} \Rightarrow 0$$



positive GM & PM



negative GM & PM

1. Sketch the bode plot for the following transfer function and determine the phase and gain cross over frequency.

$$G(s) = \frac{20}{s(1+3s)(1+4s)}$$

sinusoidal transfer function:

$$G(j\omega) = \frac{20}{j\omega(1+3j\omega)(1+4j\omega)}$$

corner frequency:

$$\omega_{c1} = \frac{1}{4} = 0.25 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{3} = 0.33 \text{ rad/sec}$$

Term	corner frequency rad/sec	slope dB/sec	change in slope dB/sec
$20 / j\omega$	—	-20	—
$1 / (1+4j\omega)$	$\omega_{c1} = 0.25$	-20	$-20 - 20 = -40$
$1 / (1+3j\omega)$	$\omega_{c2} = 0.33$	-20	$-40 - 20 = -60$

magnitude plot:

$$\omega_l < \omega_{c1} \Rightarrow \omega_l = 0.1 \text{ rad/sec}$$

$$\omega_h > \omega_{c2} \Rightarrow \omega_h = 1 \text{ rad/sec}$$

$$A = |G(j\omega)| \text{ in dB}$$

$$\text{at } \omega = \omega_l \Rightarrow A = 20 \log \left| \frac{20}{0.1} \right| = 46 \text{ dB}$$

$$\text{at } \omega = \omega_{c1} \Rightarrow A = 20 \log \left| \frac{20}{0.25} \right| = 38 \text{ dB}$$

$$\text{at } \omega = \omega_{c2} \Rightarrow A = \left| \text{slope from } \omega_{c1} \text{ to } \omega_{c2} * \log \frac{\omega_{c2}}{\omega_{c1}} \right| +$$

$$A \text{ at } \omega = \omega_{c1}$$

$$= \left[-40 * \log \frac{0.33}{0.25} \right] + 38$$

$$A = 33 \text{ dB}$$

$$\text{at } \omega = \omega_h \Rightarrow A = \left| \text{slope from } \omega_{c2} \text{ to } \omega_h * \log \frac{\omega_h}{\omega_{c2}} \right| +$$

$$A \text{ at } \omega = \omega_{c2}$$

$$= \left[-60 * \log \frac{1}{0.33} \right] + 33$$

$$A = 4 \text{ dB}$$

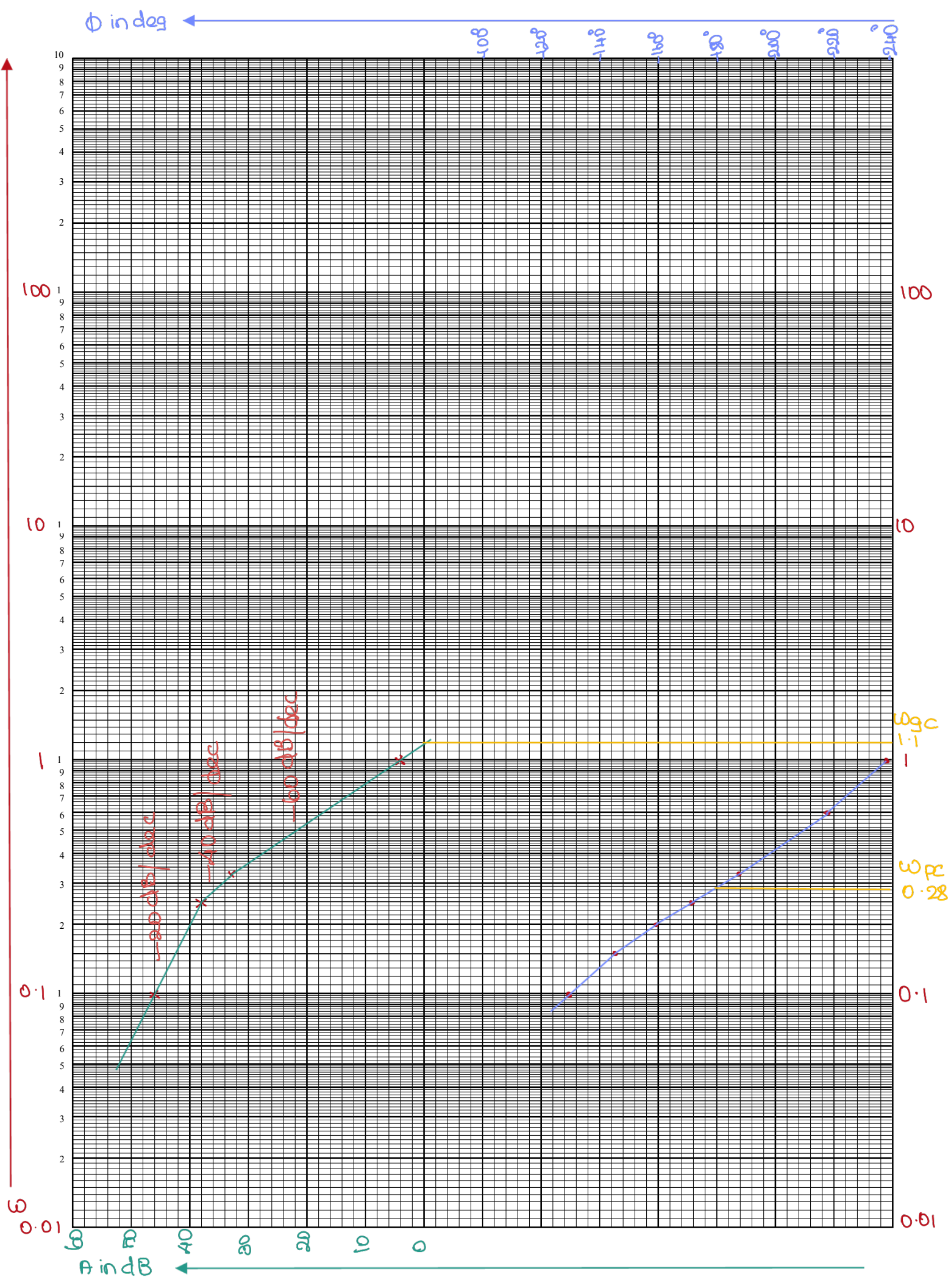
phase plot:

$$G(j\omega) = -90 - \tan^{-1}\left(-\frac{4\omega}{1}\right) - \tan^{-1}\left(-\frac{3\omega}{1}\right)$$

ω rad/sec	$\phi = \angle G(j\omega)$ deg
0.1	-129°
0.15	-146°
0.2	-160°
0.25	-172°
0.33	-188°
0.6	-218°
1	-238°

$$\omega_{gc} = 1.1 \text{ rad/sec}$$

$$\omega_{pc} = 0.28 \text{ rad/sec}$$



2. Sketch the bode plot for the following transfer function and determine the phase margin and gain margin.

$$G(s) = \frac{10}{s(1+0.4s)(1+0.1s)}$$

sinusoidal transfer function:

$$G(j\omega) = \frac{10}{j\omega(1+0.4j\omega)(1+0.1j\omega)}$$

corner frequency:

$$\omega_{c1} = \frac{1}{0.4} = 2.5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{0.1} = 10 \text{ rad/sec}$$

Term	corner frequency rad/sec	slope dB/sec	change in slope dB/sec
$10 / j\omega$	—	-20	—
$1 / (1+0.4j\omega)$	$\omega_{c1} = 2.5$	-20	$-20 - 20 = -40$
$1 / (1+0.1j\omega)$	$\omega_{c2} = 10$	-20	$-40 - 20 = -60$

magnitude plot:

$$\omega_l < \omega_{c1} \Rightarrow \omega_l = 1 \text{ rad/sec}$$

$$\omega_h > \omega_{c2} \Rightarrow \omega_h = 100 \text{ rad/sec}$$

$$A = |G(j\omega)| \text{ in dB}$$

$$\text{at } \omega = \omega_l \Rightarrow A = 20 \log \left| \frac{10}{1} \right| = 20 \text{ dB}$$

$$\text{at } \omega = \omega_{c1} \Rightarrow A = 20 \log \left| \frac{10}{2.5} \right| = 12 \text{ dB}$$

$$\text{at } \omega = \omega_{c2} \Rightarrow A = \left| \text{slope from } \omega_{c1} \text{ to } \omega_{c2} * \log \frac{\omega_{c2}}{\omega_{c1}} \right| +$$

$$A \text{ at } \omega = \omega_{c1}$$

$$= \left[-40 * \log \frac{10}{2.5} \right] + 12$$

$$A = -12 \text{ dB}$$

$$\text{at } \omega = \omega_h \Rightarrow A = \left| \text{slope from } \omega_{c2} \text{ to } \omega_h * \log \frac{\omega_h}{\omega_{c2}} \right| +$$

$$A \text{ at } \omega = \omega_{c2}$$

$$= \left[-60 * \log \frac{100}{10} \right] + (-12)$$

$$A = -72 \text{ dB}$$

phase plot:

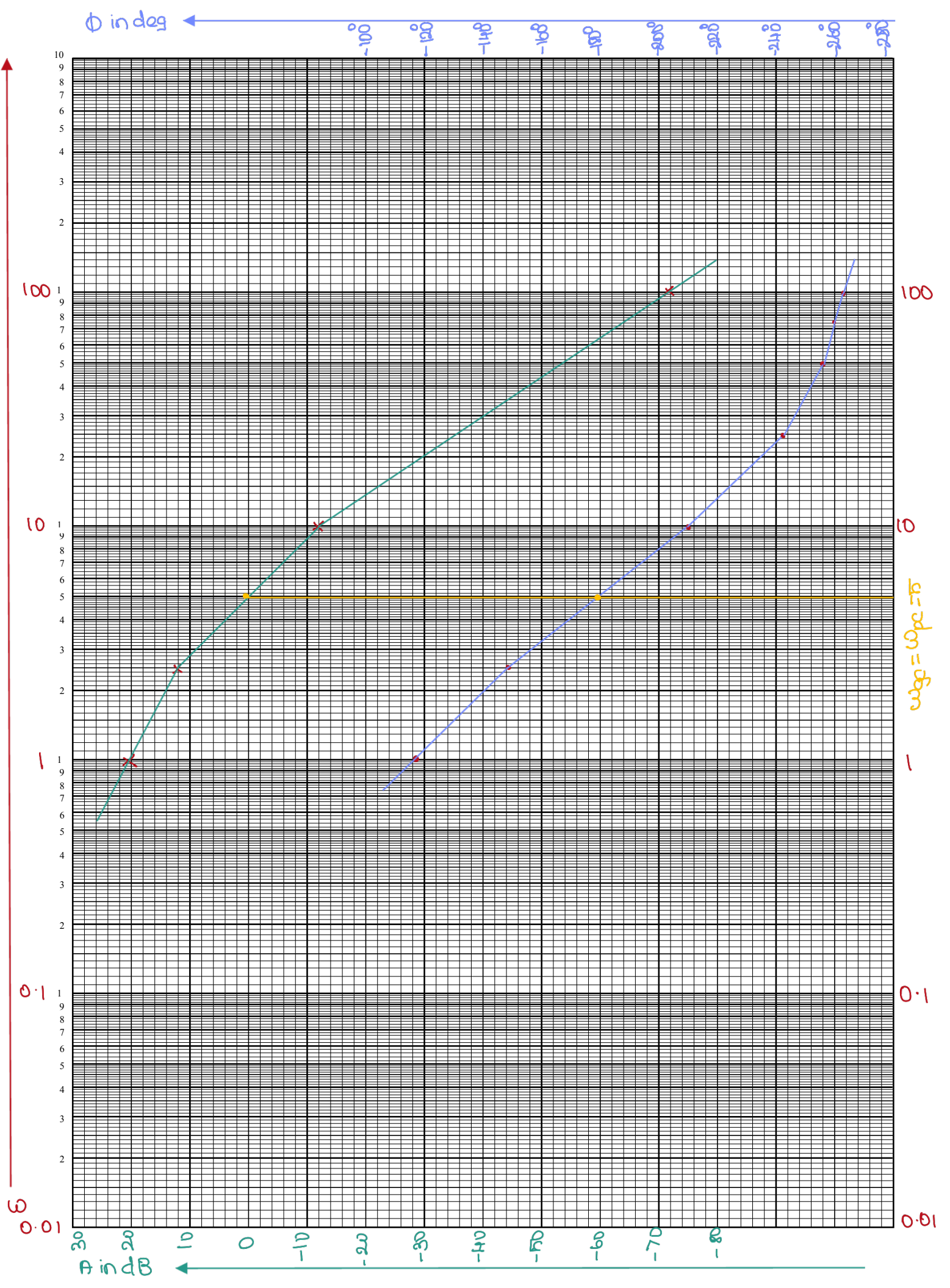
$$G(j\omega) = -90 - \tan^{-1} \left(\frac{0.4\omega}{1} \right) - \tan^{-1} \left(\frac{0.1\omega}{1} \right)$$

ω rad/sec	$\phi = \angle G(j\omega)$ deg
1	-118°
2.5	-149°
10	-210°
25	-242°
50	-256°
75	-260°
100	-263°

$$\omega_{gc} = \omega_{pc} = 5 \text{ rad/sec}$$

$$\text{GM} \quad k_g = 0 \text{ dB} \quad \text{at } \omega_{pc} = 5 \text{ rad/sec}$$

$$\text{PM} \quad \nu = 180^\circ + \phi_{gc} = 180^\circ - 180^\circ = 0^\circ$$



3. Sketch the bode plot for the following transfer function and determine the phase margin and gain margin.

$$G(s) = \frac{5}{s (1+0.2s) (1+0.02s)}$$

4. Sketch the bode plot for the following transfer function and determine the phase margin and gain margin.

$$G(s) = \frac{5 (1+2s)}{(1+4s) (1+0.25s)}$$

5. Sketch the bode plot for the following transfer function and determine the phase margin and gain margin.

$$G(s) = \frac{75 (1+0.2s)}{s (s^2+16s+100)}$$

Frequency Response Analysis

→ response of the system for the sinusoidal input.

→ time domain $\Rightarrow$ s-domain $\Rightarrow$ frequency domain

$$T(s) \xRightarrow[\text{L.T.}]{} T(s) \xRightarrow[\text{s} \rightarrow j\omega]{} T(j\omega)$$

→ magnitude and phase of sinusoidal transfer function $\Rightarrow$ function of frequency ω .

↳ open loop transfer function:

$$G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

↳ loop transfer function:

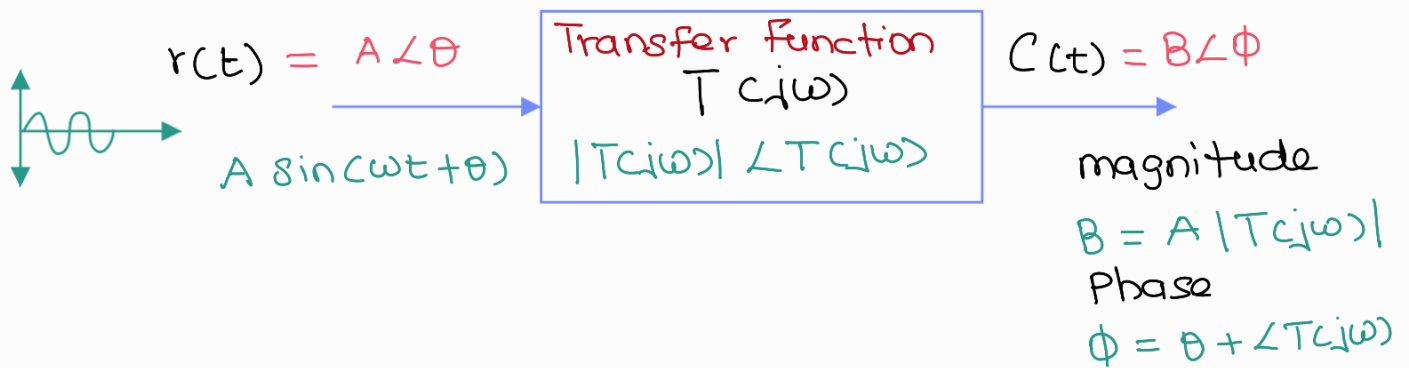
$$G(j\omega)H(j\omega) = |G(j\omega)H(j\omega)| \angle G(j\omega)H(j\omega)$$

↳ unity feedback system:

$$\begin{aligned} H(j\omega) &= 1 \\ \Rightarrow G(j\omega)H(j\omega) &= G(j\omega) \end{aligned}$$

↳ closed loop transfer function:

$$\frac{G(j\omega)}{1+G(j\omega)H(j\omega)} = T(j\omega) = |T(j\omega)| \angle T(j\omega)$$



→ merits:

- ↳ absolute and relative stability of the closed loop system can be estimated from the knowledge of their open loop transfer function.
- ↳ practical testing of system can be easily carried with available sinusoidal signal generators and precise measurement equipments.
- ↳ transfer function of complicated systems can be determined experimentally by frequency response tests.
- ↳ design and parameter adjustment of the open loop transfer function of a system for specified closed loop performance is carried out more easily in frequency domain.
- ↳ effects of noise disturbance and parameter variations are relatively easy to visualize and incorporate corrective measures.
- ↳ can be extended to certain non-linear control systems.

Frequency domain specifications

→ resonant peak:

↳ maximum value of the magnitude of closed loop transfer function

large resonant peak in frequency response



large overshoot in transient response

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

→ resonant frequency:

↳ frequency at which the resonant peak occurs.

resonant frequency in frequency response



frequency of oscillation in step response

$$\omega_r = \omega_n \sqrt{1-2\zeta^2}$$

→ cut-off frequency:

↳ frequency at which the gain of the system is -3dB.

→ bandwidth:

↳ range of frequencies for which normalized gain of the system is more than -3dB.

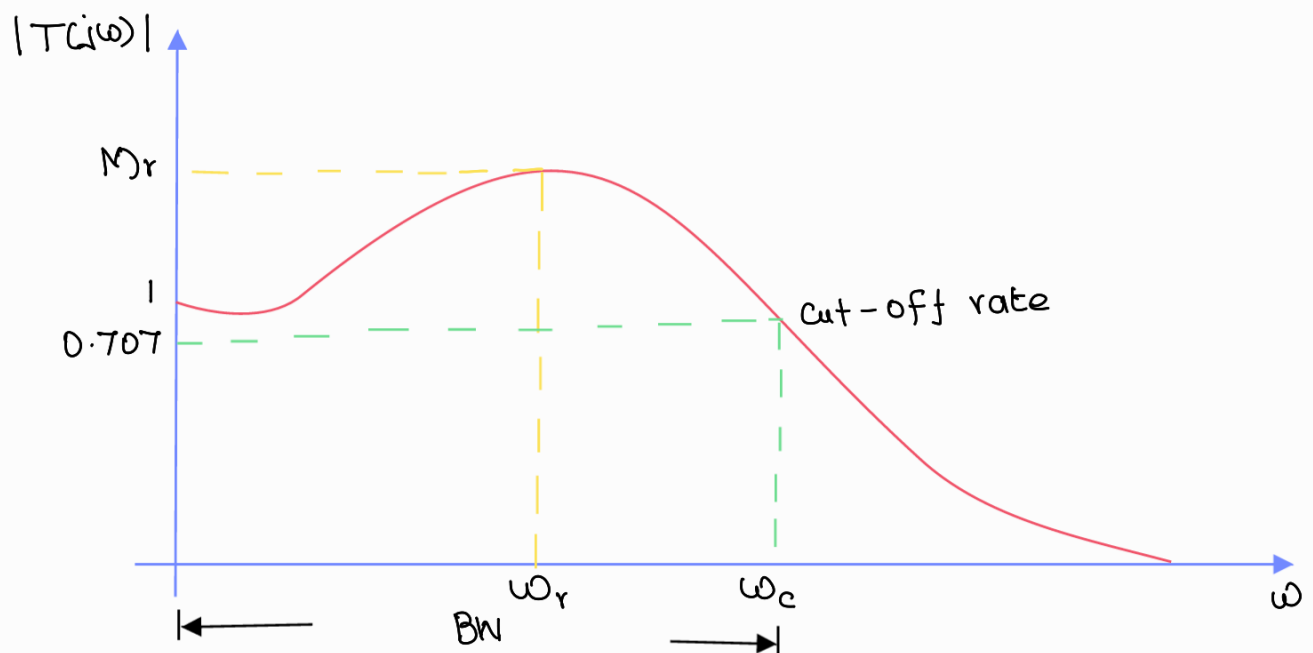
↳ measure of the ability of a feedback system to reproduce the input, noise rejection characteristics and rise time.

large bandwidth in frequency response



small rise time or fast response

$$\omega_b = \omega_n \left[1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4} \right]^{1/2}$$



→ cut-off rate:

↳ slope of the log-magnitude curve near the cut-off frequency.

↳ ability of the system to distinguish the signal from noise.

→ gain cross over frequency:

↳ frequency at which the magnitude of the open loop transfer function is unity (dB magnitude $\Rightarrow 0$).

$$\omega_{gc} \Rightarrow |G(j\omega)| = 1$$

→ phase cross over frequency:

↳ frequency at which the phase of the open loop transfer function is 180° .

$$\omega_{pc} \Rightarrow \angle G(j\omega) = 180^\circ$$

→ gain margin:

↳ value of gain to be added/provided to the system without affecting the stability.

↳ given by reciprocal of the magnitude of open loop transfer function at phase cross over frequency.

$$K_g = \frac{1}{|G(j\omega_{pc})|} \quad \text{at } \omega = \omega_{pc}$$

$$\text{in dB} \Rightarrow K_g = 20 \log \frac{1}{|G(j\omega_{pc})|} = -20 \log |G(j\omega_{pc})|$$

→ phase margin:

↳ additional phase lag to be added/provided to the system without affecting the stability.

↳ given by adding 180° to the phase angle of open loop transfer function at gain cross over frequency.

$$\gamma = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = \angle G(j\omega_{gc}) \quad \text{at } \omega = \omega_{gc}$$

correlation between frequency domain and time domain specifications

→ for second order system:

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

↳ undamped natural frequency $\Rightarrow \omega_n$

↳ damping ratio $\Rightarrow \zeta$

→ in frequency domain:

↳ $s \Rightarrow j\omega$

↳ resonant peak $\Rightarrow$

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

↳ resonant frequency $\Rightarrow$

$$\omega_r = \omega_n \sqrt{1-2\zeta^2}$$

→ in time domain:

↳ peak overshoot $\Rightarrow$

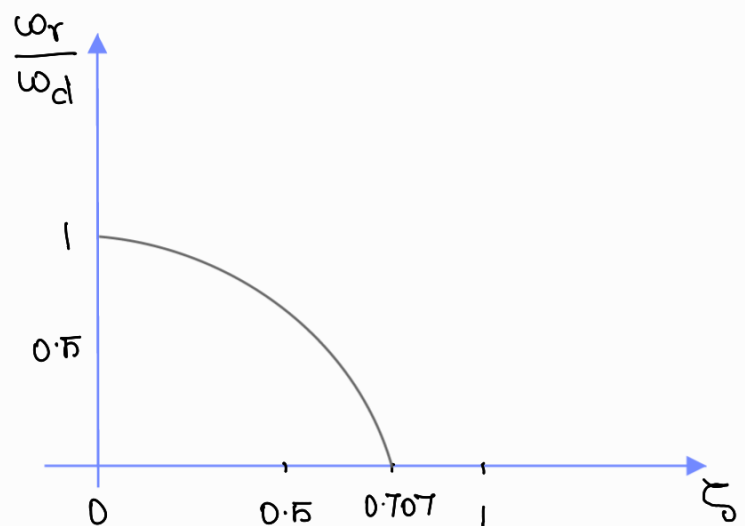
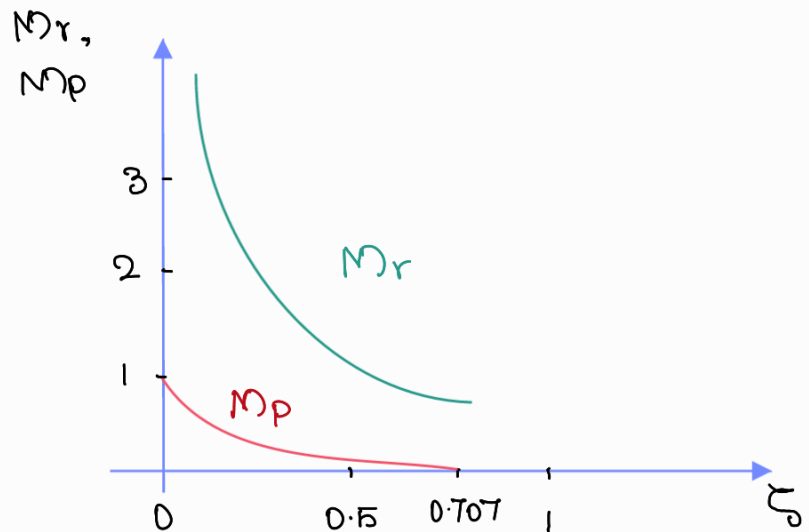
$$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

↳ damped frequency $\Rightarrow$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

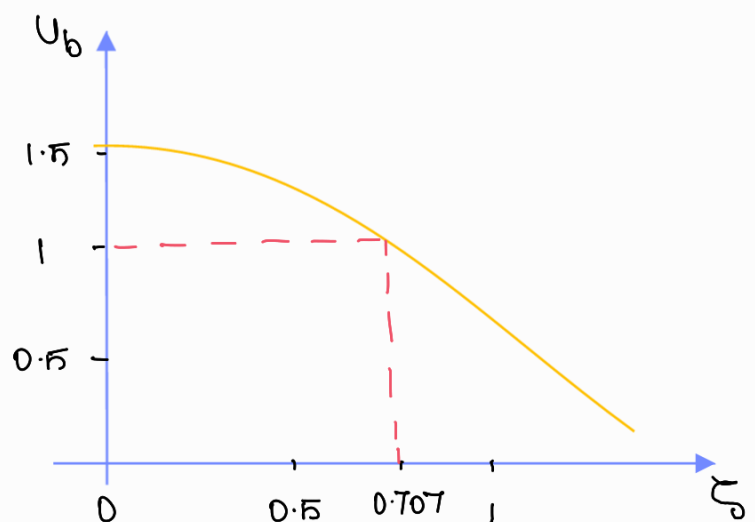
→ M_r, M_p, ω_r and ω_d are function of ζ_0 .

→ comparison:



→ normalised bandwidth as function of ζ_0 :

$$u_b = \frac{\omega_b}{\omega_n} = [1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + \zeta^4}]^{1/2}$$



↳ frequency domain specifications can be expressed in terms of time domain parameters ζ and ω_n

large resonant peak in frequency response
 $\Rightarrow$ $\Updownarrow$
large overshoot in transient response

resonant frequency in frequency response
 $\Rightarrow$ $\Updownarrow$
frequency of oscillation in step response

large bandwidth in frequency response
 $\Rightarrow$ $\Updownarrow$
small rise time or fast response

1. The damping ratio and natural frequency of oscillation of a second order system is 0.5 and 8 rad/sec respectively. Calculate the resonant peak and resonant frequency.

↳ resonant peak $\Rightarrow M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$

$$= \frac{1}{2 * (0.5) \sqrt{1 - 0.5^2}}$$
$$M_r = 1.154$$

↳ resonant frequency $\Rightarrow \omega_r = \omega_n \sqrt{1 - 2\zeta^2}$

$$= 8 * \sqrt{1 - (2 * (0.5)^2)}$$
$$\omega_r = 5.66 \text{ rad/sec}$$

Frequency Response plots:

- ↳ Bode plot
- ↳ Polar/ Nyquist plot
- ↳ Nichols plot
- ↳ M & N circles